

Auditing Semantic Gains in Sequential Recommendation: A Lightweight Recovery Test

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Abstract

Recent semantic and generative-retrieval recommenders report substantial improvements over ID-only sequential baselines. However, it remains unclear whether these gains arise from language-model reasoning, semantic-ID generation, end-to-end semantic architectures, stronger offline item representations, or the complementarity of semantic and collaborative signals. We investigate this attribution ambiguity through LIME-Rec, a lightweight and auditable recovery test. LIME-Rec combines three independent experts: a SASRec sequential expert, an ItemCF co-occurrence expert, and a semantic expert based on frozen BAAI/bge-base-en-v1.5 item embeddings. Each expert produces full-catalog scores, which are normalized per user and combined through auditable score-level fusion, followed by bounded history calibration. The fusion gate and calibration head are fitted using validation data only, require no serving-time language-model inference, and keep each expert contribution separately inspectable. On the text-rich Amazon Beauty, Toys, and Sports benchmarks, LIME-Rec achieves R@10 scores of 0.0996, 0.1105, and 0.0593, exceeding the strongest comparison method in the table by 7.0%–12.0%. A factorial isolation ablation further shows that three-expert fusion without history calibration consistently outperforms calibrated SASRec, ruling out bounded history calibration as the sole explanation for the recovery. Randomly permuting item-text embeddings across item IDs while preserving model capacity reduces R@10 by 13.6%–17.5%, showing that the gains depend on genuine item-text correspondence rather than additional representation capacity alone. These results do not imply that complex semantic models are universally unnecessary. Instead, they show that lightweight recovery from offline item representations and transparent fusion should be ruled out before improvements are attributed to serving-time language modeling, semantic-ID generation, or heavier semantic machinery. Code is available at <https://github.com/Double-wk/LIME-Rec>.

Introduction

Sequential recommendation has traditionally focused on modeling user preference dynamics from historical interaction sequences. Early sequential recommenders based on recurrent and convolutional architectures demonstrated that temporal patterns in user behaviors provide valuable signals beyond static collaborative filtering (Hidasi et al. 2016;

Tang and Wang 2018). With the development of Transformer architectures, self-attentive models such as SASRec and BERT4Rec further improved sequential recommendation by capturing complex dependencies in user histories (Kang and McAuley 2018; Sun et al. 2019). Despite their effectiveness, these ID-based approaches mainly rely on user-item interaction signals and represent items through discrete identifiers, limiting their ability to exploit rich semantic information associated with items.

Recent research has therefore extended sequential recommendation beyond ID-based modeling by incorporating semantic information. One direction introduces item-side textual information and pretrained semantic representations to enhance item understanding and improve recommendation quality (Hou et al. 2022; Li et al. 2023). Another direction explores large language models (LLMs) for recommendation, leveraging their language understanding and reasoning capabilities to interpret user histories and recommendation tasks (Geng et al. 2022; Bao et al. 2023; Liao et al. 2024). Meanwhile, generative retrieval methods incorporate semantic identifiers into the retrieval process itself, replacing conventional item ranking with semantic item generation (Rajput et al. 2023; Zheng et al. 2024). Together, these studies demonstrate that semantic information can provide valuable signals beyond traditional collaborative interactions and have led to substantial improvements on text-rich recommendation benchmarks.

However, despite these advances, the source of semantic recommendation gains remains unclear. Existing semantic and generative recommenders are typically evaluated as complete systems, making it difficult to determine which factors actually contribute to their improvements over ID-only baselines. The observed gains may come from sophisticated language-model reasoning, semantic retrieval mechanisms, stronger item representations derived from text, or simply complementary effects between semantic and collaborative signals. These possibilities lead to different conclusions about whether increasingly complex recommendation architectures are truly necessary. If the main benefit comes from online language-model inference or semantic generation, then additional architectural complexity may be justified. However, if a substantial portion of the improvement can already be recovered through lightweight offline representations and simple fusion, then the contribution of complex se-

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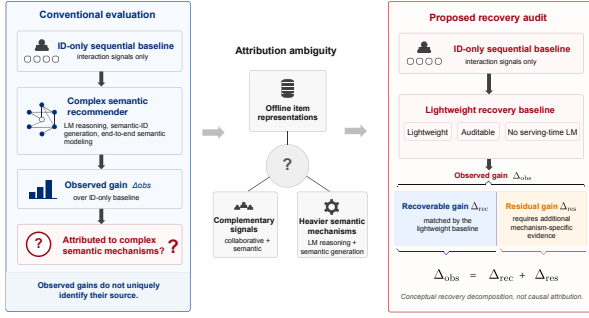


Figure 1: Motivation for LIME-Rec. Performance gains over an ID-only baseline do not uniquely identify their source, motivating a lightweight recovery audit.

semantic mechanisms should be reconsidered. We refer to this issue as *attribution ambiguity*: improvements over ID-only baselines do not uniquely reveal their underlying sources.

As illustrated in Figure 1, performance gains over an ID-only sequential baseline do not uniquely identify whether the improvement originates from lightweight offline signals or heavier semantic mechanisms. To investigate this problem, we propose **LIME-Rec**, a lightweight recovery test for auditing semantic gains in sequential recommendation. Rather than introducing another complex semantic architecture, LIME-Rec evaluates whether reported semantic improvements can already be explained by simple, transparent, and offline components. Experiments on Amazon Beauty, Toys, and Sports show that LIME-Rec recovers substantial semantic recommendation gains, achieving relative R@10 improvements of 7.0%–12.0%. Controlled analyses further show that these gains rely on genuine item-text correspondence rather than additional representation capacity, and a factorial isolation of multi-expert fusion and bounded history calibration rules out history calibration as the sole explanation for the recovery.

Our contributions are summarized as follows:

1. We identify *attribution ambiguity* in semantic sequential recommendation, showing that improvements over ID-only baselines do not uniquely reveal their underlying sources.
2. We propose **LIME-Rec**, a lightweight recovery test that provides an auditable reference for evaluating whether semantic recommendation gains require complex semantic architectures.
3. We conduct controlled experiments on three text-rich Amazon benchmarks, including a factorial isolation of multi-expert fusion and bounded history calibration, item-text correspondence controls, and expert-complementarity diagnostics that separate semantic-information effects from architectural and post-processing effects.

Related Work

Sequential recommendation. Sequential recommendation models learn user preference dynamics from historical

interaction sequences. Early recurrent and convolutional approaches demonstrate the effectiveness of modeling temporal dependencies in user behaviors (Hidasi et al. 2016; Tang and Wang 2018). With the development of Transformer architectures, self-attentive models such as SASRec and BERT4Rec further improve sequence representation learning by capturing long-range dependencies in user histories (Kang and McAuley 2018; Sun et al. 2019). Subsequent studies introduce self-supervised and contrastive objectives to improve robustness under sparse behavioral signals (Zhou et al. 2020; Xie et al. 2022). These ID-based sequential recommenders provide the collaborative foundation for our study, but improvements over them alone cannot determine whether additional semantic information provides independent recommendation value.

Semantic-enhanced recommendation. A line of research incorporates item semantics into sequential recommendation through textual information and pretrained representations. UniSRec learns transferable sequence representations by leveraging item-side information, while RecFormer exploits language representations of item content for sequential recommendation (Hou et al. 2022; Li et al. 2023). More generally, pretrained text encoders make it practical to transform item descriptions into semantic representations for downstream recommendation tasks (Reimers and Gurevych 2019). These approaches demonstrate the value of item semantics, but their end-to-end optimization jointly involves text encoding, sequential modeling, and recommendation objectives, making it difficult to isolate how much improvement originates from item representations themselves.

LLM-based and generative recommendation. Recent studies explore large language models (LLMs) for recommendation by reformulating recommendation tasks as language understanding or generation problems. P5 introduces a unified text-to-text recommendation framework, while TALLRec and LLaRA investigate adapting LLMs to personalized recommendation scenarios through instruction tuning and lightweight alignment strategies (Geng et al. 2022; Bao et al. 2023; Liao et al. 2024). Another research direction integrates semantics into the retrieval process itself through semantic identifiers and generative retrieval. TIGER introduces semantic IDs for generative recommendation, while LC-Rec integrates collaborative semantics into an LLM-based recommendation framework (Rajput et al. 2023; Zheng et al. 2024). Although these approaches achieve strong performance, their tightly coupled architectures make it challenging to attribute observed gains to specific components, such as language understanding, semantic representations, or retrieval mechanisms.

Hybrid recommendation and score fusion. Hybrid recommendation combines collaborative and content-based evidence through feature combination, cascaded ranking, or score fusion strategies (Burke 2002; Adomavicius and Tuzhilin 2005). Existing hybrid methods mainly optimize predictive accuracy, often relying on learned fusion mechanisms that make individual evidence sources increasingly difficult to separate. In contrast, LIME-Rec intentionally

adopts transparent score-level fusion, where different evidence sources remain independently observable. This design enables controlled analysis of how much recommendation improvement can be explained by individual components.

Position of LIME-Rec. Different from existing semantic recommenders and hybrid systems, LIME-Rec is not designed as another accuracy-oriented recommendation architecture. Instead, it serves as a lightweight recovery test for auditing semantic gains. By evaluating whether improvements can already be recovered through simple and auditable offline components, LIME-Rec provides an attribution reference before assigning gains to complex language modeling, semantic generation, or learned retrieval mechanisms.

Method

Recovery Test Overview

LIME-Rec is designed as a lightweight recovery test for auditing semantic gains in sequential recommendation rather than as another accuracy-oriented recommendation architecture. Its goal is to measure how much of the improvement attributed to semantic or generative recommendation can already be recovered using simple, transparent, and independently inspectable components.

As illustrated in Figure 2, LIME-Rec separates one-time semantic representation construction from online recommendation. Item text is encoded offline into frozen item embeddings, while three independent experts produce sequential, collaborative, and semantic scores at recommendation time. These full-catalog scores are normalized per user, combined through auditable score-level fusion, adjusted by bounded history calibration, and finally used for full-catalog Top- K ranking.

Audit interpretation. LIME-Rec is a mechanism-necessity test rather than a causal decomposition of a target model. If a lightweight system that excludes serving-time language modeling, semantic-ID decoding, and end-to-end text optimization reaches the performance of a heavier method under the same data and evaluation protocol, those excluded mechanisms are not empirically necessary to attain that benchmark score. This result does not imply that the heavier model internally performs the same computation or that its mechanisms are unhelpful in other settings. Instead, the test quantifies the performance that remains attainable after removing those mechanisms from the serving path while retaining offline item semantics and standard interaction signals.

Let $\mathcal{U}$ and $\mathcal{I}$ denote the user and item sets. For each user $u \in \mathcal{U}$, the chronological interaction history is

$$H(u) = (i_1, i_2, \dots, i_{T_u}).$$

The three independent experts produce full-catalog score vectors

$$s_x(u, \cdot) \in \mathbb{R}^{|\mathcal{I}|}, \quad x \in \{\text{seq}, \text{cf}, \text{sem}\},$$

where the indices denote the sequential, collaborative, and semantic experts, respectively. The experts are trained or constructed independently, do not share parameters or internal representations, and interact only through their full-catalog score vectors and statistics derived from those vectors.

Independent Experts

Sequential Expert We use SASRec (Kang and McAuley 2018) as the sequential expert. Given the chronological history $H(u)$, SASRec models user preference dynamics and produces a score for every candidate item $i \in \mathcal{I}$:

$$s_{\text{seq}}(u, i).$$

This expert provides sequential behavioral evidence without using item textual information.

Collaborative Expert We construct an ItemCF expert to provide explicit collaborative co-occurrence evidence. Let $H_L(u)$ denote the suffix containing the most recent L interactions, ordered from oldest to newest within the suffix:

$$H_L(u) = (h_1, h_2, \dots, h_L), \quad L = \min(20, |H(u)|).$$

The collaborative score for candidate item i is

$$s_{\text{cf}}(u, i) = \sum_{t=1}^L \frac{t}{L} \frac{\text{cooc}(h_t, i)}{\sqrt{N_{h_t} N_i}}.$$

Here, $\text{cooc}(h_t, i)$ is the co-occurrence count between items h_t and i , and N_i is the frequency of item i in the training interaction sequences. The factor t/L assigns greater weight to recent interactions, while the square-root normalization attenuates raw popularity effects. All co-occurrence counts and item frequencies are computed exclusively from the training interactions.

Semantic Expert The semantic expert provides item-text evidence through frozen offline representations. For each item, the available title, brand, category, and description fields are concatenated and encoded once using the frozen English BGE checkpoint BAAI/bge-base-en-v1.5 from the BGE/C-Pack project (Xiao et al. 2024).

Let $e_i \in \mathbb{R}^d$ denote the ℓ_2 -normalized embedding of item i . We assign a recency weight to each historical item:

$$r_j = \exp[-\gamma(T_u - j)], \quad \gamma = 0.1.$$

The semantic user representation is computed as the normalized recency-weighted average of the historical item embeddings:

$$\bar{e}_u = \text{normalize} \left(\frac{\sum_{j=1}^{T_u} r_j e_{i_j}}{\sum_{j=1}^{T_u} r_j} \right).$$

The semantic relevance score for candidate item i is

$$s_{\text{sem}}(u, i) = \langle \bar{e}_u, e_i \rangle.$$

The text encoder is used only for offline item-embedding construction. At recommendation time, the semantic expert operates on stored embeddings and performs no text encoding, generation, or language-model inference.

Per-User Score Normalization

The three experts produce scores on different numerical scales. LIME-Rec therefore applies min-max normalization separately for each user and expert over the full item catalog:

$$\tilde{s}_x(u, i) = \frac{s_x(u, i) - \min_{j \in \mathcal{I}} s_x(u, j)}{\max_{j \in \mathcal{I}} s_x(u, j) - \min_{j \in \mathcal{I}} s_x(u, j) + \epsilon}.$$

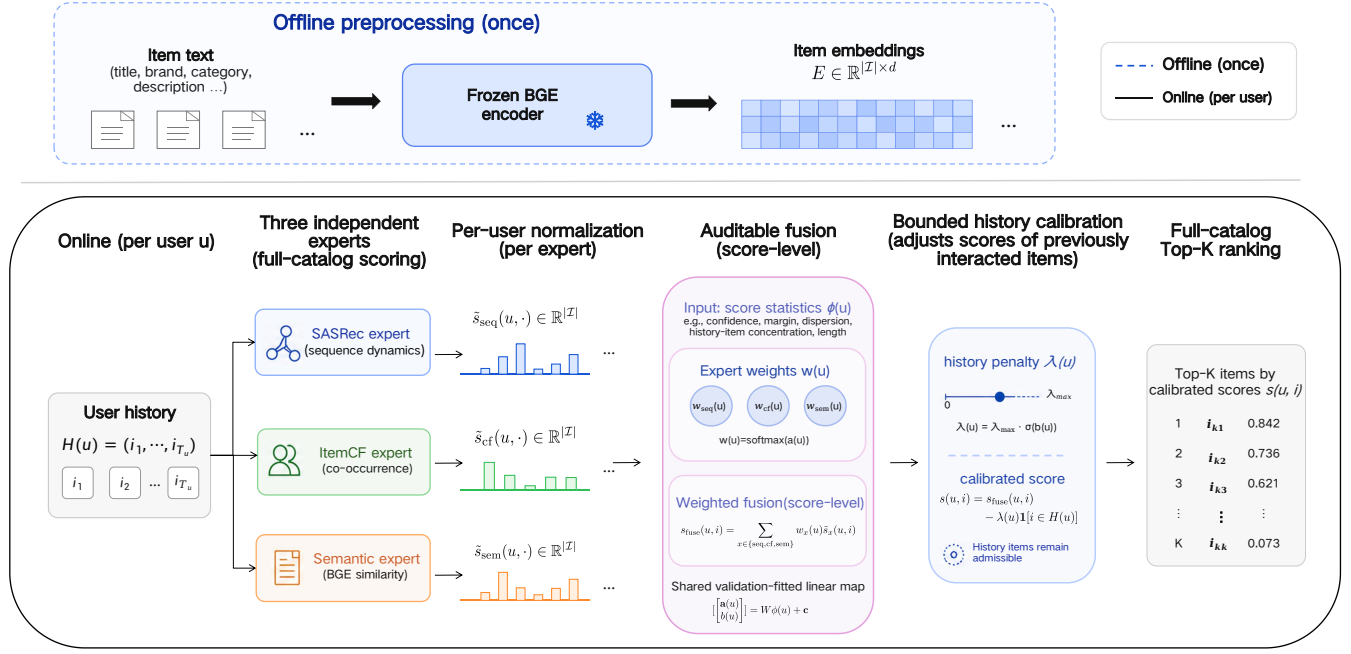


Figure 2: Overview of LIME-Rec. Frozen item embeddings are constructed offline, while three independent experts produce full-catalog scores for per-user normalization, auditable fusion, bounded history calibration, and Top- K ranking.

Here, $x \in \{\text{seq}, \text{cf}, \text{sem}\}$ and $\epsilon > 0$ is a small constant used for numerical stability. This monotonic transformation preserves each expert’s within-user ranking while placing the three full-catalog score vectors on comparable numerical scales.

Auditable Score-Level Fusion

LIME-Rec combines the normalized expert scores through a lightweight per-user fusion gate. Let $\phi(u)$ denote an inference-time feature vector computed from the normalized score distributions. It contains the user’s history length and, for each expert, summary statistics describing score confidence, top-score margin, score dispersion, and the concentration of previously interacted items among high-ranked candidates.

A lightweight linear map produces three expert-weight logits $\mathbf{a}(u) \in \mathbb{R}^3$ and one calibration logit $b(u) \in \mathbb{R}$:

$$\begin{bmatrix} \mathbf{a}(u) \\ b(u) \end{bmatrix} = W\phi(u) + \mathbf{c}.$$

The expert-weight vector is obtained through a softmax transformation:

$$\mathbf{w}(u) = \text{softmax}(\mathbf{a}(u)) = (w_{\text{seq}}(u), w_{\text{cf}}(u), w_{\text{sem}}(u)).$$

The fused score is

$$s_{\text{fuse}}(u, i) = \sum_{x \in \{\text{seq}, \text{cf}, \text{sem}\}} w_x(u) \tilde{s}_x(u, i).$$

The fusion gate consumes only summary statistics derived from the expert score distributions, while the weighted sum

operates on the complete normalized full-catalog score vectors. It therefore introduces no interaction between the experts’ internal representations, and each expert score and assigned weight remains separately inspectable.

Bounded History Calibration

After auditable score-level fusion, LIME-Rec applies bounded history calibration to adjust the fused scores of previously interacted items:

$$s(u, i) = s_{\text{fuse}}(u, i) - \lambda(u) \mathbb{I}[i \in H(u)].$$

The calibration magnitude is obtained from the calibration logit $b(u)$ and constrained by a predefined upper bound $\lambda_{\max}$:

$$\lambda(u) = \lambda_{\max} \sigma(b(u)), \quad 0 \leq \lambda(u) \leq \lambda_{\max}.$$

We use $\lambda_{\max} = 0.10$ in all reported experiments. Auditable fusion and bounded history calibration serve different roles. The former combines complementary evidence from the three experts, whereas the latter adjusts the fused scores of previously interacted items. Although their logits are produced as separate outputs of the same lightweight linear map over $\phi(u)$, they act on different parts of the scoring pipeline. The Results section therefore evaluates them as separate experimental factors through independently fitted ablations.

Serving Path

LIME-Rec separates one-time semantic preprocessing from online recommendation. During offline preprocessing, item text is converted into frozen item embeddings:

$$\text{item text} \rightarrow \text{frozen text encoder} \rightarrow \text{item embeddings}.$$

At recommendation time, the online path consists of SASRec scoring, ItemCF score accumulation, similarity computation over stored item embeddings, per-user score normalization, auditable score-level fusion, bounded history calibration, and full-catalog Top- K selection.

Thus, the serving path requires no online text encoding, semantic-ID generation, or autoregressive decoding. The pretrained language encoder is used only for one-time offline item-representation construction.

Experimental Setup

Datasets and evaluation protocol. We conduct experiments on the standard 5-core Amazon Reviews 2014 (McAuley et al. 2015) Beauty, Toys, and Sports datasets; detailed statistics are reported in Table S1. For each user, the chronologically latest interaction is held out for test, the second latest for validation, and all earlier interactions form the training history. At test time, we rank the full catalog with `mask_history=false`, so previously interacted items remain eligible candidates; we report Recall (R) and NDCG (N) at cutoffs 5 and 10. Reported confidence intervals come from a paired user-level bootstrap on the seed-0 run and quantify within-run sampling uncertainty rather than seed-to-seed variation (details in the Results section).

Baselines. We compare LIME-Rec with representative methods spanning semantic recommendation, language-model-enhanced recommendation, and generative retrieval, including TIGER (Rajput et al. 2023), IDGenRec (Tan et al. 2024), HSTU (Zhai et al. 2024), LC-Rec (Zheng et al. 2024), GRAM (Lee et al. 2025), and MHL (Wei et al. 2026).

Implementation details. For each dataset and seed $s \in \{0, 1, 2\}$ we independently train one SASRec checkpoint and reuse it across every configuration that includes the sequential expert, so differences reflect expert composition rather than backbone initialization. The ItemCF and semantic experts are constructed as described in the Method section; the full architecture, optimization, and fusion settings are listed in Table S7. The gate and calibration head are fitted on validation interactions only; the test split is not used for parameter

fitting, checkpoint selection, early stopping, or hyperparameter selection, and is accessed solely for final evaluation.

Results

Main recovery results. Table 1 reports the main full-catalog recommendation results. LIME-Rec achieves the highest performance in all twelve dataset–metric combinations, showing that its gains are consistent across datasets, ranking cutoffs, and evaluation metrics. Relative to GRAM, the strongest baseline in every reported column, LIME-Rec improves R@5 by 9.1%, 9.5%, and 8.4% on Beauty, Toys, and Sports, respectively. The corresponding improvements on R@10 are 12.0%, 12.0%, and 7.0%. NDCG also improves consistently at both cutoffs, with relative gains ranging from 8.6% to 10.6%.

The simultaneous improvements in Recall and NDCG indicate that LIME-Rec not only retrieves more relevant target items within the top- K list, but also tends to place them at higher ranks. Although all methods obtain lower absolute scores on Sports than on Beauty and Toys, LIME-Rec maintains a positive margin over the strongest baseline under every Sports metric. Overall, these results show that competitive semantic-recommendation performance can be recovered through frozen item-text representations, standard sequential and collaborative signals, and lightweight score-level fusion.

Stability and paired uncertainty analysis. Across the three training seeds, LIME-Rec achieves R@10 scores of 0.0996 ± 0.0009 , 0.1105 ± 0.0005 , and 0.0593 ± 0.0008 on Beauty, Toys, and Sports, respectively, where the variation denotes the sample standard deviation across seeds $\{0, 1, 2\}$. The corresponding improvements over the matched SAS-Rec backbone, computed before rounding, are $+0.0235$, $+0.0359$, and $+0.0176$. These gains are substantially larger than the observed seed-to-seed variation on all three datasets, indicating stable improvements across independently trained runs.

Using the seed-0 predictions, a paired user-level bootstrap with 1,000 resamples yields LIME-Rec-minus-SASRec

Model	Beauty				Toys				Sports			
	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10
TIGER (Rajput et al. 2023)	0.0352	0.0236	0.0533	0.0294	0.0274	0.0174	0.0438	0.0227	0.0176	0.0143	0.0311	0.0146
IDGenRec (Tan et al. 2024)	0.0463	0.0328	0.0665	0.0393	0.0462	0.0323	0.0651	0.0383	0.0273	0.0186	0.0403	0.0228
HSTU (Zhai et al. 2024)	0.0469	0.0314	0.0704	0.0389	0.0433	0.0281	0.0669	0.0357	0.0258	0.0165	0.0414	0.0215
LC-Rec (Zheng et al. 2024)	0.0503	0.0352	0.0715	0.0420	0.0543	0.0385	0.0753	0.0453	0.0259	0.0175	0.0384	0.0216
GRAM (Lee et al. 2025)	0.0641	0.0451	0.0890	0.0531	0.0718	0.0516	0.0987	0.0603	0.0375	0.0256	0.0554	0.0314
MHL (Wei et al. 2026)	0.0574	0.0424	0.0795	0.0495	0.0672	0.0489	0.0903	0.0564	0.0359	0.0249	0.0511	0.0298
LIME-Rec	0.0699	0.0491	0.0996	0.0587	0.0786	0.0561	0.1105	0.0664	0.0407	0.0281	0.0593	0.0341

Table 1: Main recovery results on the Amazon Beauty, Toys, and Sports datasets. LIME-Rec results are averaged over three random seeds, $\{0, 1, 2\}$.

Table 2: Factorial ablation of three-expert fusion and bounded history calibration. R@10 is reported as mean $\pm$ sample standard deviation over seeds $\{0, 1, 2\}$ under the full-catalog, repeat-allowed protocol.

Dataset	SASRec	SASRec + Cal.	Fusion w/o Cal.	Full LIME-Rec
Beauty	0.0761 $\pm$ 0.0023	0.0810 $\pm$ 0.0024	0.0882 $\pm$ 0.0009	0.0996 $\pm$ 0.0009
Toys	0.0746 $\pm$ 0.0008	0.0789 $\pm$ 0.0006	0.0991 $\pm$ 0.0003	0.1105 $\pm$ 0.0005
Sports	0.0417 $\pm$ 0.0011	0.0439 $\pm$ 0.0009	0.0519 $\pm$ 0.0008	0.0593 $\pm$ 0.0008

R@10 improvements of +0.0238 with a 95% confidence interval of $[+0.0213, +0.0267]$ on Beauty, +0.0351 with $[+0.0315, +0.0387]$ on Toys, and +0.0194 with $[+0.0172, +0.0217]$ on Sports. All three intervals exclude zero, supporting a positive average paired improvement at the user level. The bootstrap point estimates differ slightly from the differences between the three-seed means because the bootstrap analysis uses paired seed-0 predictions, whereas the reported means average independently trained runs.

Isolating expert fusion from history calibration. Because the main evaluation retains previously interacted items as eligible candidates, we examine whether the improvement over the matched SASRec backbone can be explained primarily by bounded history calibration. We independently fit four variants on the validation split: SASRec without calibration, SASRec with calibration, three-expert fusion without calibration, and the complete LIME-Rec model. Disabled components are removed throughout validation fitting and evaluation rather than suppressed only after fitting. All variants reuse the same matched SASRec checkpoint for each dataset-seed pair and follow the same full-catalog, repeat-allowed evaluation protocol.

Calibration alone yields relatively modest R@10 improvements of +0.0048, +0.0043, and +0.0022 over SASRec on Beauty, Toys, and Sports, respectively. In contrast, three-expert fusion without calibration improves R@10 by +0.0120, +0.0244, and +0.0102. Therefore, bounded history calibration is not sufficient to explain the recovery over the matched SASRec backbone: multi-expert fusion remains substantially beneficial when the history adjustment is disabled.

Adding calibration to the fused scores provides a further gain of +0.0115, +0.0115, and +0.0074 on the three datasets. The conditional calibration gain is larger after fusion than when applied to SASRec alone, suggesting positive complementarity between the two components rather than a simple duplication of the same effect. Fusion supplies additional sequential, co-occurrence, and semantic evidence, while calibration refines the treatment of previously interacted items in the final ranking.

Additional robustness checks show that replacing per-user calibration with a global scalar or BGE with MiniLM produces only minor changes in R@10; complete results are reported in Table S6.

Attribution Analysis

We assess the source of the recovered gains along three dimensions: whether performance depends on correct item-text alignment, whether the experts expose non-redundant candidate regions, and whether combining these signals produces measurable improvements over the individual experts.

Item-text correspondence control. Figure 3 tests whether the semantic contribution depends on the correct correspondence between items and their textual representations. Randomly permuting the frozen text embeddings across item IDs preserves the number and dimensionality of the embeddings, the semantic scoring architecture, the fusion module, and the full-catalog evaluation procedure, while removing meaningful item-text alignment. Under this intervention, R@10 decreases from 0.0996 to 0.0851 on Beauty, from 0.1105 to 0.0912 on Toys, and from 0.0593 to 0.0512 on Sports, corresponding to relative drops of 14.6%, 17.5%, and 13.6%.

The fusion gate changes in the same direction. The mean semantic-expert weight decreases from 0.318 to 0.204 on Beauty, from 0.405 to 0.216 on Toys, and from 0.329 to 0.205 on Sports. The joint reduction in ranking performance and semantic weight is consistent with the gain depending on meaningful item-text correspondence rather than on the mere addition of another score channel.

Target-level complementarity. Table S2 decomposes target retrieval according to the exact subset of singleton experts

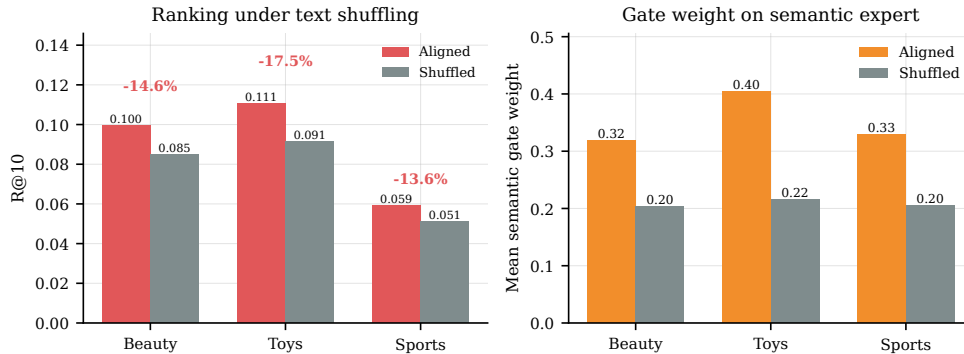


Figure 3: Item-text correspondence control. Shuffling item-text embeddings across item IDs lowers R@10 (left, with relative drop) and reduces the gate’s mean semantic-expert weight (right) on all three datasets.

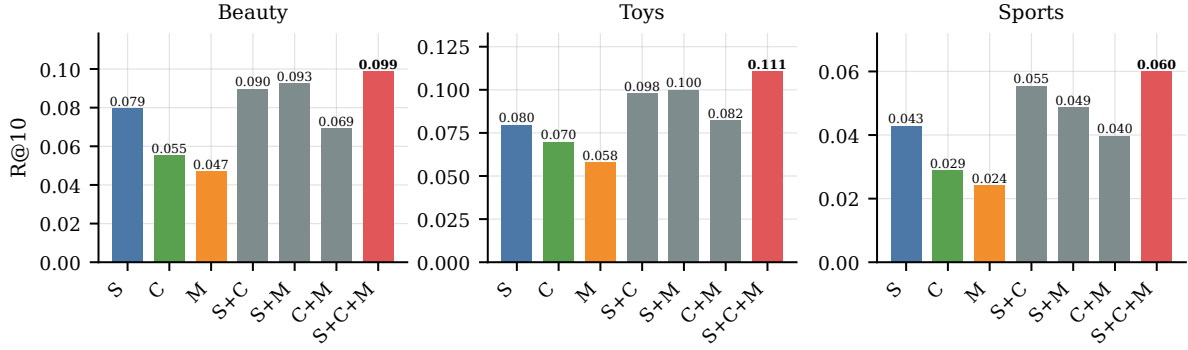


Figure 4: Expert ablation on seed 0 (S: SASRec; C: ItemCF; M: Semantic). Full metrics appear in Table S3.

that places the ground-truth item in its Top-10. The semantic expert uniquely retrieves the target for 2.65%, 2.55%, and 1.40% of the sampled Beauty, Toys, and Sports users, respectively. These semantic-only hit rates are larger than the corresponding ItemCF-only rates of 1.65%, 2.05%, and 1.25%, showing that semantic evidence reaches relevant candidate regions missed by both interaction-based experts.

The pairwise categories further show that expert support is only partially overlapping. ItemCF and Semantic jointly retrieve targets missed by SASRec for 0.40%, 0.60%, and 0.35% of users on Beauty, Toys, and Sports, respectively, whereas targets retrieved by all three experts account for only 0.95%, 1.80%, and 0.15%. Together with the low Jaccard overlap and limited rank correlations in Table S5, these results show that the experts provide distinct and target-relevant recommendation evidence.

Contribution of individual experts. Figure 4 connects candidate-level non-redundancy to measurable recommendation gains. Although the semantic expert is the weakest singleton, adding it to SASRec improves R@5, N@5, R@10, and N@10 on all three datasets. For example, SASRec + Semantic improves R@10 from 0.0794 to 0.0926 on Beauty, from 0.0795 to 0.1000 on Toys, and from 0.0429 to 0.0487 on Sports. The full three-expert fusion further outperforms every singleton and pairwise configuration across all twelve reported metrics. Full R@5, N@5, and N@10 results are reported in Table S3.

These results show that singleton strength alone does not determine an expert’s value within the fused system. The semantic expert contributes a weaker but non-redundant signal, while the sequential and co-occurrence experts provide stronger interaction-based evidence. Combining all three sources produces the most effective final ranking.

Taken together, the item–text correspondence control, candidate-overlap diagnostics, target-hit decomposition, and singleton and pairwise ablations show that correctly aligned item text provides useful evidence that is not redundant with sequential or co-occurrence signals, and that lightweight fusion converts this complementary evidence into measurable recommendation gains. The supplementary material provides a finer-grained decomposition of the recovered users on Beauty.

Scope and Validity

Our conclusions are limited to the three text-rich Amazon benchmarks considered here. In a preliminary Yelp probe, reported separately because its structured attributes differ from Amazon product text and it was not included in the full attribution battery, fusion with frozen MiniLM embeddings (Yelp Inc. 2026) improves the three-seed mean R@10 from 0.0495 ± 0.0003 to 0.0596 ± 0.0008 ; the semantic expert receives a mean gate weight of 0.251, below the 0.32–0.41 range on Amazon. This probe does not establish a text-sparsity boundary or cross-domain generalization.

LIME-Rec requires informative item content and a transferable encoder, but no serving-time text encoding, language-model inference, or semantic-ID decoding. Its factorial isolation applies to the repeat-allowed protocol; the component decomposition may differ when previously interacted items are excluded. Heavier semantic architectures should therefore be evaluated beyond this recovery baseline.

Conclusion

We introduced LIME-Rec, a lightweight recovery test for auditing semantic gains in sequential recommendation. On three text-rich Amazon benchmarks, LIME-Rec combines SASRec, ItemCF, and frozen BAAI/bge-base-en-v1.5 item embeddings through validation-only score-level fusion and bounded history calibration, recovering substantial semantic recommendation gains without serving-time language-model inference. The fusion–calibration isolation, singleton and pairwise ablations, candidate-overlap diagnostics, and shuffled-embedding control jointly indicate that the recovery cannot be explained by repeat-aware history calibration alone: multi-source expert fusion provides the primary standalone gain, while bounded calibration supplies a complementary improvement to the final ranking. Future work should compare against this recovery baseline before attributing gains to serving-time language modeling, semantic-ID generation, or heavier architectures.

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Dataset Statistics

Table S1 summarizes the interaction data. Beauty, Toys, and Sports are the three Amazon datasets used for all main experiments and controlled attribution analyses. Yelp is included only as a **Yelp preliminary probe**; it is not part of the main Amazon evaluation suite or the full attribution battery.

Dataset	Users	Items	Interactions
Beauty	22,363	12,101	198,502
Toys	19,412	11,924	167,597
Sports	35,598	18,357	296,337
Yelp (prelim. probe)	30,431	20,033	316,354

Table S1: Dataset statistics. Yelp is a preliminary probe only.

Additional Attribution Details

We analyze users recovered beyond the matched SASRec backbone.

Recovered-User Decomposition

For Beauty seed 0, fusion recovers 64 users whose ground-truth targets are absent from the matched SASRec Top-10. Of these users, 21 (32.8%) are pure rank-promotion cases: no singleton expert retrieves the target in its Top-10, but fusion does. Among the remaining recovered users, 26.6% are CF-only, 29.7% are semantic-only, and 10.9% are joint CF-semantic singleton hits.

These cases show that fusion improves recommendation through two related mechanisms. Some targets are recovered directly because another expert retrieves candidates missed by SASRec, whereas others are promoted into the Top-10 by combining multiple individually insufficient score signals.

Full Ablation Metrics

This section reports the complete singleton and pairwise ablation summarized by Figure 4 in the main paper. Table S3 lists Recall and NDCG at cutoffs 5 and 10 for all seven expert configurations under the same full-catalog protocol. Across every dataset and metric, the three-expert fusion achieves the best result.

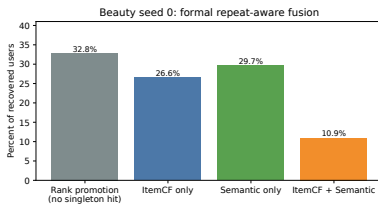


Figure S1: Recovered-user attribution for Beauty, seed 0.

Each pairwise configuration that includes the semantic expert improves over the corresponding sequential or collaborative singleton, confirming that the semantic signal contributes non-redundant evidence beyond R@10.

Additional Experiments and Diagnostics

Original-SASRec Control

Table S4 repeats fusion with the original SASRec setup—one-negative BCE and the ICDM’18-style block—rather than the main full-vocabulary CE and pre-normalized block. R@10 improves from 0.0473 to 0.0817 on Beauty, 0.0550 to 0.0999 on Toys, and 0.0227 to 0.0516 on Sports (+72.9%, +81.4%, and +127.2%). Recovery therefore also holds in the original setup, although absolute scores are not directly comparable.

Candidate-Level Non-Redundancy

Table S5 shows consistently low top-10 overlap and weak-to-moderate full-catalog rank association between the three experts. The Jaccard@10 values range from 0.019 to 0.081, indicating that the experts frequently expose different candidates in their highest-ranked regions. Their full-catalog Spearman correlations range from 0.080 to 0.238, showing that these differences extend beyond only the top-ranked items.

Additional Robustness Checks

Calibration parameterization and encoder choice. Table S6 examines whether the recovered performance depends strongly on the calibration parameterization or frozen text encoder. Replacing the per-user calibration output with a single global calibration scalar changes R@10 by at most 0.0007. Replacing BGE with MiniLM under the same SASRec checkpoints, fusion procedure, candidate catalog, and evaluation protocol changes R@10 by at most approximately 0.0011. These results indicate that the recovery is not tied to individualized calibration or a particular frozen text encoder.

Gate Construction and Training Details

For each user and expert e , we first min-max normalize the full-catalog score vector independently:

$$\tilde{s}_e(u, i) = \frac{s_e(u, i) - \min_j s_e(u, j)}{\max_j s_e(u, j) - \min_j s_e(u, j) + 10^{-8}}. \quad (1)$$

The additive constant also defines the behavior for a constant score vector, which is mapped to zero. The gate input $\phi(u) \in \mathbb{R}^{14}$ concatenates $\log(1 + |H_u|)/5$; four statistics for each of the three normalized expert vectors (top-1 score minus the catalog mean, the top-1-top-2 margin, catalog standard deviation, and the fraction of that expert’s Top-5 items contained in H_u); and the fraction of catalog items contained in H_u .

A single linear layer maps $\phi(u)$ to four logits. Softmax over the first three produces the expert weights, while $\lambda(u) = 0.10 \text{ sigmoid}(z_4)$ produces the bounded history penalty. The two outputs are trained jointly on validation interactions by minimizing

$$\mathcal{L} = \text{CE} \left(20 \left[\sum_e w_e(u) \tilde{s}_e(u, \cdot) - \lambda(u) \mathbf{1}[\cdot \in H_u] \right], y_u \right). \quad (2)$$

Dataset	NONE	only-SAS	only-CF	only-Sem	SAS+CF	SAS+Sem	CF+Sem	ALL
Beauty	87.45	4.35	1.65	2.65	2.30	0.25	0.40	0.95
Toys	86.55	4.20	2.05	2.55	1.95	0.30	0.60	1.80
Sports	93.10	2.95	1.25	1.40	0.75	0.05	0.35	0.15

Table S2: Top-10 target-hit decomposition for 2,000 fixed seed-0 test users (sampling seed 42). Entries are percentages; NONE and ALL denote no expert and all three experts.

Configuration	Beauty				Toys				Sports			
	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10
SASRec only	0.0582	0.0414	0.0794	0.0483	0.0584	0.0420	0.0795	0.0489	0.0286	0.0194	0.0429	0.0240
ItemCF only	0.0375	0.0251	0.0554	0.0308	0.0495	0.0330	0.0698	0.0395	0.0191	0.0136	0.0290	0.0167
Semantic only	0.0297	0.0199	0.0469	0.0254	0.0377	0.0243	0.0579	0.0307	0.0147	0.0098	0.0241	0.0128
SASRec + ItemCF	0.0650	0.0466	0.0896	0.0545	0.0719	0.0529	0.0977	0.0613	0.0388	0.0269	0.0555	0.0323
SASRec + Semantic	0.0647	0.0447	0.0926	0.0536	0.0704	0.0494	0.1000	0.0590	0.0314	0.0207	0.0487	0.0262
ItemCF + Semantic	0.0449	0.0290	0.0693	0.0369	0.0549	0.0380	0.0821	0.0468	0.0255	0.0176	0.0397	0.0222
Three experts	0.0692	0.0491	0.0986	0.0586	0.0794	0.0566	0.1106	0.0666	0.0412	0.0285	0.0599	0.0345

Table S3: Seed-0 singleton and pairwise expert-composition ablations on Amazon Beauty, Toys, and Sports. All configurations retain bounded history calibration; Table 2 in the main paper isolates that factor. R and N denote Recall and NDCG.

Dataset	SASRec	Fusion	Δ R@10
Beauty	0.0473 $\pm$ 0.0008	0.0817 $\pm$ 0.0021	+72.9%
Toys	0.0550 $\pm$ 0.0022	0.0999 $\pm$ 0.0005	+81.4%
Sports	0.0227 $\pm$ 0.0004	0.0516 $\pm$ 0.0012	+127.2%

Table S4: Original-SASRec control (one-negative BCE and the ICDM’18-style block). Values are three-seed mean $\pm$ sample standard deviation; gains use unrounded means.

Dataset	Jaccard@10			Full-catalog Spearman		
	S-C	S-M	C-M	S-C	S-M	C-M
Beauty	0.0572	0.0401	0.0635	+0.1065	+0.1092	+0.0816
Toys	0.0524	0.0484	0.0812	+0.0947	+0.1430	+0.0799
Sports	0.0191	0.0249	0.0472	+0.1078	+0.2382	+0.0938

Table S5: Candidate-overlap diagnostics for 2,000 fixed seed-0 test users (sampling seed 42). S, C, and M denote SASRec, ItemCF, and Semantic; Spearman uses full-catalog average ranks and Jaccard uses Top-10 lists.

Validation examples use the training history to predict the validation item; test examples append the validation item to the training history and predict the held-out test item. Test interactions never fit or select gate parameters. For the shuffled-text control, the semantic item-embedding correspondence is permuted first and the complete gate, including calibration, is then refitted from scratch on the validation split before test evaluation.

Audit Scope and Interpretation

Role of the architecture. LIME-Rec uses standard, inspectable components as a controlled recovery test rather than a new expert architecture. This design exposes excluded mechanisms and supports factorial isolation, expert ablation, and item-text correspondence controls.

Table S6: R@10 robustness to calibration and the frozen encoder. MiniLM and BGE reuse identical SASRec checkpoints; results average three seeds.

Dataset	Calibration		Encoder		LIME-Rec vs. SASRec		
	Per-user	Global	MiniLM	BGE	SASRec	LIME-Rec	Gain
Beauty	0.0996	0.0990	0.0996	0.0996	0.0761	0.0996	+0.0235
Toys	0.1105	0.1099	0.1095	0.1105	0.0746	0.1105	+0.0359
Sports	0.0593	0.0590	0.0584	0.0593	0.0417	0.0593	+0.0176

Component	Configuration
Architecture	Pre-normalized Transformer block; maximum sequence length 50; hidden size 64; 2 blocks; 2 attention heads; dropout 0.2.
Training	Full-vocabulary item-ID cross-entropy; Adam; learning rate 10^{-3} ; no weight decay; batch size 128; up to 200 epochs; validation every 10 epochs; patience 30.
Checkpointing	Independent checkpoint per dataset-seed pair; matched variants reuse it; seeds {0, 1, 2}; test used only for final evaluation.
Semantic expert	Frozen BAAI/bge-base-en-v1.5; offline, ℓ_2 -normalized item embeddings.
Fusion module	One linear layer: three expert-weight logits and one calibration logit.
Fusion training	Validation-only; initial weights (0.60, 0.15, 0.25) (sequential, collaborative, semantic); score scale 20; $\lambda_{\max} = 0.10$; AdamW; learning rate 0.02; weight decay 10^{-3} ; batch size 256; 4 epochs.
Environment	Ubuntu 22.04; 8-core CPU; 32 GB RAM; NVIDIA A10 GPU (24 GB); Python 3.12; PyTorch 2.10.0; CUDA 12.8; Transformers 4.55.4.

Table S7: Model, optimization, and fusion settings.

Interpretation of calibration. Bounded history calibration is part of LIME-Rec but not the sole source of recovery. In Table 2, fusion without calibration improves R@10 over SASRec by 0.0120, 0.0244, and 0.0102 on Beauty, Toys, and Sports; calibration adds a complementary gain. Thus the full margin reflects the complete auditable pipeline, not semantic complementarity alone.